MIL-P-4971A
20 August 1956

Superseding
MIL-P-4971 (USAF)
23 August 1955

MILITARY SPECIFICATION

PANEL, CONTROL TRANSFER, AIRFIELD LIGHTING, TYPE MB-1

This specification has been approved by the Department of Defense, and is mandatory for use by the Departments of the Army, the Navy and the Air Force.

1. SCOPE

1.1 This specification covers one type of airfield lighting, control transfer panel, designated Type MB-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

LL-B-631 — Boxes; Fiber Corrugated (for Domestic Shipment).

LLL-B-636 — Boxes; Fiber, Solid, (for Domestic Shipment).

PPP-B-585 — Boxes; Wood, Wirebound.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

MILITARY

JAN-P-100 — Packaging and Packing for Overseas Shipment, General Specification.

JAN-P-103 — Packaging and Packing for Overseas Shipment Boxes; Wood Cleated; Solid Fiberboard.

JAN-P-106 — Boxes; Wood, Nailed (Overseas Type).

JAN-P-108 — Boxes, Fiberboard, Corrugated and Solid (Military Type).

MIL-P-116 — Preservation, Methods of.

JAN-P-127 — Packaging and Packing for Overseas Shipment — Tape, Adhesive, Pressure-Sensitive, Water-Resistant.

MIL-B-138 — Boxes, Wood, Fiberboard-Lined for Overseas Shipment (for Weight Contents not Exceeding 500 Pounds).

MIL-B-4229 — Boxes; Paperboard, Metal-Stayed.

FSC 6110

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MIL-D-5028 — Drawings and Data Lists; Preparation of Manufacturers' (for Production Aircraft, Guided Missiles, Engines, Accessories, and Other Auxiliary Equipment).

MIL-L-10547 — Liners, Case, Waterproof.

STANDARDS

MILITARY

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-130 — Identification Marking of U. S. Military Property.

PUBLICATION

AIR FORCE-NAVY AERONAUTICAL BULLETINS

143 — Specifications and Standards; Use of.

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 OTHER PUBLICATION.—The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BOARD OF FIRE UNDERWRITERS

Pamphlet 70 — National Electrical Code (Electrical Wiring and Apparatus).

(Copies of the NBFU pamphlet may be obtained upon application to the National Fire Protection Association, 60 Batterymarch Street, Boston 10, Massachusetts.)

3. REQUIREMENTS

3.1 Qualification.—The panel furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein.

3.2 Components.—The panel shall consist of the following major components:

Item	*Description*	*See Requirement*
1	Enclosure	3.7
2	Mounting plate	3.8
3	Grounding lug	3.9
4	Relay	3.10
5	Terminal strips	3.11
6	Wiring	3.12

3.3 Selection of Specifications and Standards.—Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with Bulletin 143.

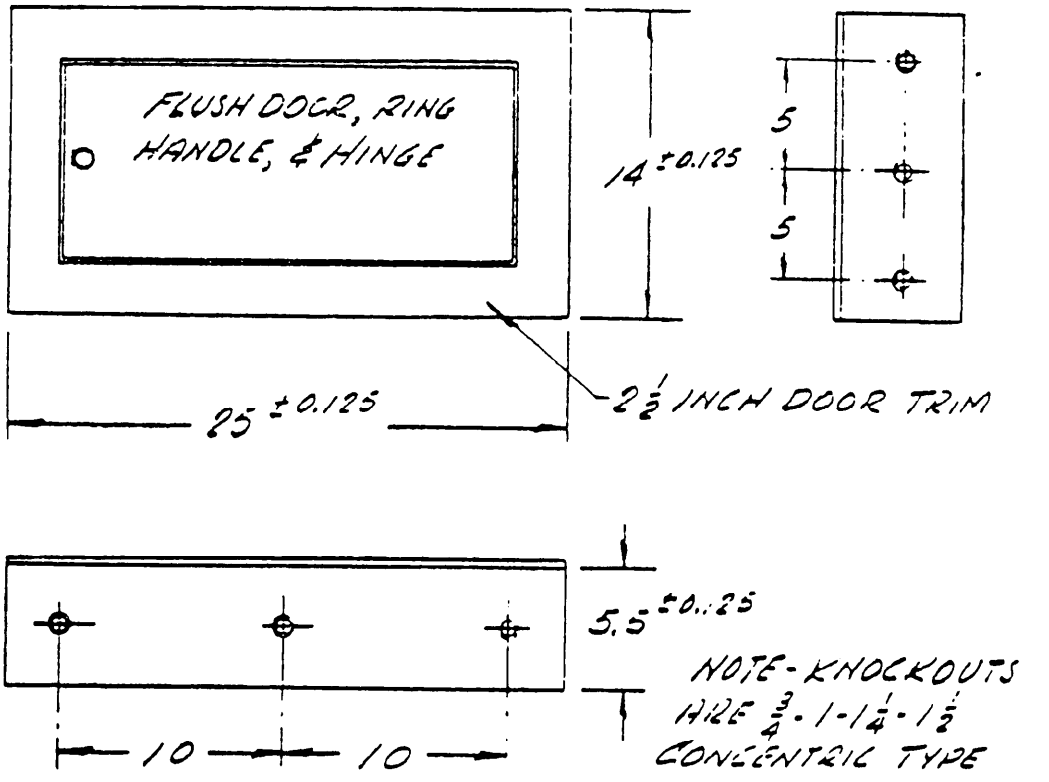
3.4 Materials.

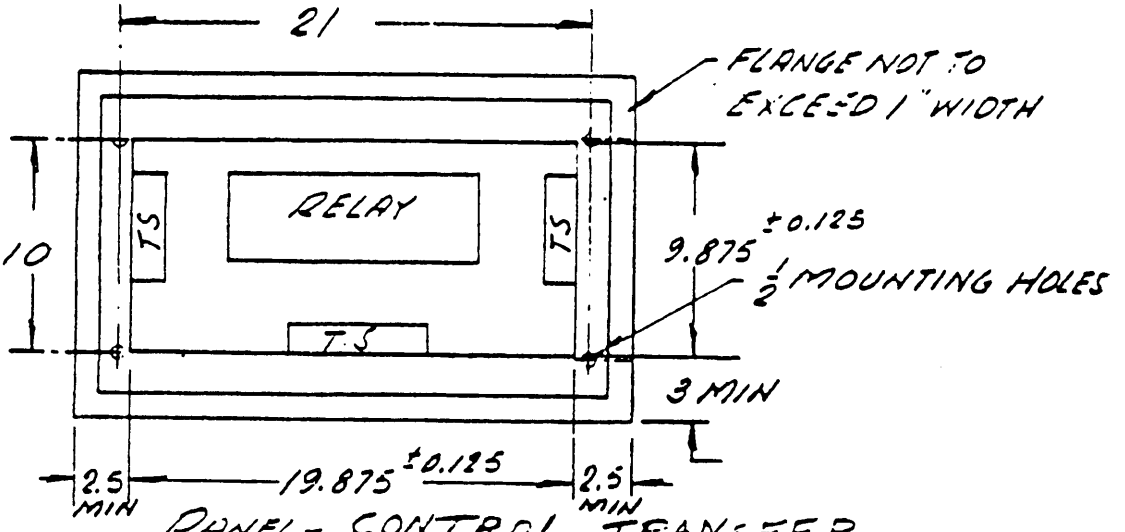
3.4.1 *Fungus-Proof Materials.*—Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.5 Design.—The panel shall be designed for indoor use in accordance with high-grade commercial practice and shall meet the applicable requirements of Pamphlet 70 of the National Electrical Code.

3.6 Construction.—The panel shall be designed and constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

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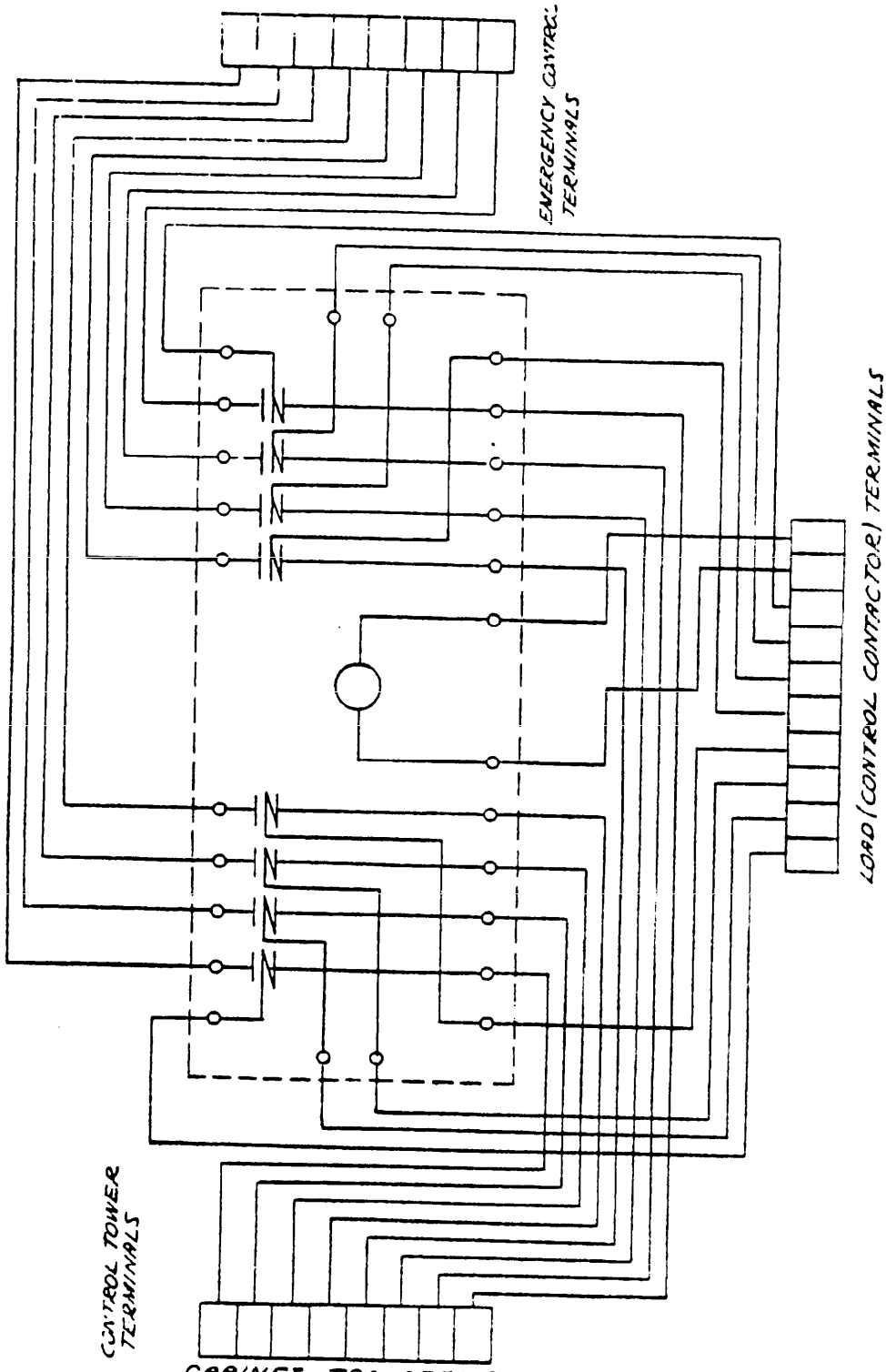




PANEL - CONTROL TRANSFER, AIRPORT LIGHTING, TYPE MB-1
OUTLINE DIMENSIONS & PANEL ARRANGEMENTS

DIMENSIONS IN INCHES FIGURE 1

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CABINET - TRANSFER RELAY PANEL WIRING DIAGRAM (F.V.)

FIGURE 2

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3.7 Enclosure.—The panel shall have a NEMA type 1 enclosure. The enclosure shall be completely rigid, self-supporting, and designed for surface or rack mounting.

3.7.1 The enclosure, including the door and trim, shall be fabricated of 0.109 ± 0.020 inch sheet steel. The enclosure shall have a black, satin, corrosion-resistant finish.

3.7.2 *Dimensions.*—The dimensions of the enclosure shall be as specified on figure 1. Dimensions and tolerances not specified shall be consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability of component parts, they shall be held or limited accordingly.

3.7.3 *Knockouts.*—Three each ¾-, 1-, 1¼, and 1½-inch concentric knockouts shall be provided on the top, bottom, and each side of the enclosure. The knockouts on each side shall be symmetrically positioned on 5-inch centers. The knockouts on the top and bottom shall be on 10-inch centers.

3.7.4 *Door and Trim.*—The door trim shall be of one piece and shall be 2½ inches wide. The trim shall be held to the flange provided on the enclosure by sufficient trim clamps to insure rigidity and a tight fit between the trim and enclosure. The door shall be flush. Hardware for the door shall consist of flush hinges and a flush spring latch with a ring handle. The hinge shall be mounted vertically on the right side of the door. A neat molding shall completely surround the door in order to provide reinforcement and a rabbet around the door opening. The molding shall be mitered around the corners.

3.7.5 *Mounting Holes.* — Four ½-inch mounting holes shall be provided as shown on figure 1.

3.8 Mounting Plate.—A removable, 0.125 ± 0.020 inch steel mounting plate shall be provided for mounting the relay and terminal strips. The dimensions of the plate shall be as specified on figure 1. The plate shall be drilled and tapped to facilitate mounting the relay and terminal strip. The steel plate shall be mounted ¼ inch from the back of the cabinet and shall be positioned as specified on figure 1. The relay and terminal strips shall be arranged on the steel plate to provide a wiring gutter of not less than 2½ inches at the bottom. The general arrangement of the relay and terminal strips shall be as specified on figure 1.

3.9 Grounding Lug.—A grounding lug, capable of accommodating a size 8 AWG wire, shall be provided inside the enclosure.

3.10 Relay.—An eight-pole, double-throw, continuous-duty, standard-commercial, heavy-duty, industrial-control-type relay shall be mounted on the plate specified in 3.8. The physical arrangement of the operating coil and terminals shall be as shown on figure 2. The normal position of the relay base shall be vertical with the normally closed contacts on the lower side of the relay. The relay operating coil shall be rated at 115V, 60 cycles. The continuous holding current shall not exceed 0.35 amp at 115V, 60 cycles. The relay shall pull in at 90V or less and shall not drop out at voltage exceeding 65V. The contacts shall have a rating of not less than 10 amp for continuous noninductive loads. The make-break rupture current rating of both normally open and normally closed contacts shall be at least 45 amp resistive load at 250V, 60 cycles. The air gap between contacts on any double-throw pole shall be at least 0.075 inch in the open position when the opposite contact just makes. The minimum center-to-center spacing of adjacent terminals and of adjacent stationary contacts shall be 0.75 inch and the minimum surface creepage shall be not less than 0.50 inch. Moving parts of the relay shall be removable from the front. The moving contacts shall be spring loaded in the closed position to insure positive contact and wiping action. The spring-loaded element shall not be a load-carrying part of the relay. The stationary contacts shall be rigidly mounted. The relay

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terminals for external connection shall be screw type; solder-type terminals shall not be acceptable.

3.11 **Terminal Strips.** — Standard-commercial terminal strips shall be mounted on the plate specified in 3.8. The terminals shall be rated at not less than 20 amp, and shall accommodate wire up to No. 8 AWG. Adequate space for the installation of incoming wires shall be allowed between the terminal strips and sides and bottom of the enclosure. The load and coil terminal strip shall be located below the relay, and shall include not less than 10 terminals. The control tower terminal strip shall include eight terminals and shall be located to the left of the relay. The emergency control terminal strip shall contain eight terminals, and shall be located to the right of the relay. The terms left and right refer to the operator facing the panel.

3.11.1 *Marking.*—Marking strips shall be provided so that the terminals of the terminal strips may be labeled during installation. In addition, the mounting plate adjacent to each strip shall be labeled as follows:

Left terminal strip — CONTROL TOWER TERMINALS

Right terminal strip — EMERGENCY CONTROL TERMINALS

Lower terminal strip — LOAD CONTROL CONTACTOR TERMINALS

3.12 **Wiring.**—The panel shall be wired in accordance with good commercial practice. All wire shall be flame-retardent, synthetic, stranded, 600V, No. 14 AWG (National Electric Code Type T). Connections shall be as shown on figure 2. When the relay is de-energized, each control tower terminal shall be connected to the corresponding load terminal, and all emergency control terminals shall be disconnected. Similarly, when the relay is energized, each emergency control shall be connected to the corresponding load terminal, and all control tower terminals shall be disconnected.

3.12.1 *Wiring Diagram.*—A 6- by 8-inch plastic card or metal plate shall be mounted on the back of the enclosure door. The card or plate shall have a wiring diagram of the internal connections of the terminal strips and relay engraved upon it. The wiring diagram shall be similar to the diagram shown on figure 2. The wiring diagram shall be identified and placed in such a position as to facilitate its use.

3.13 **Interchangeability.**—All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL–D–5028 shall govern changes in the manufacturer's part numbers.

3.14 **Identification of Product.**—Equipment, assemblies, and parts shall be marked for identification in accordance with MIL–STD–130.

3.15 **Workmanship.**—The panel, including all parts and accessories, shall be fabricated and finished in a workmanlike manner in accordance with sound practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Classification of Tests.**—The inspection and testing of the panel shall be classified as follows:

(a) Acceptance tests—(See 4.2.)

(b) Qualification testing—(See 4.5.)

4.2 **Acceptance Tests.**—Acceptance tests shall consist of the individual tests.

4.2.1 *Individual Tests.*—Each panel shall be subjected to the following tests as described under 4.4:

(a) Examination of product

(b) Continuity

(c) Operational

(d) Dielectric

4.3 **Test Conditions.**

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4.3.1 Unless otherwise specified, the tests specified herein shall be conducted at room temperature.

4.3.2 The electrical tests specified herein shall be conducted using a suitable 60-cycle sine wave power source.

4.4 Test Methods.

4.4.1 *Examination of Product.*—The panel shall be inspected to determine compliance with requirements specified herein with respect to design, materials, workmanship, and marking.

4.4.2 *Continuity.*—The panel shall be subjected to electrical continuity tests to determine that all wiring connections are correct and that all contacts close properly.

4.4.3 *Operational.*—The panel shall be tested for proper operation of each component and for suitability of the complete panel for the intended application.

4.4.4 *Dielectric.*—The panel shall be subjected to a dielectric test conducted at 1,500V ac, or 2,000V dc to determine that each terminal, and the wires connected thereto, is properly insulated from ground and from all other wires.

4.4.5 *Contact Drop.*—The voltage across relay contacts as measured at the relay terminals shall be determined at rated continuous resistive load through the contacts and minimum coil voltage. The drop shall not exceed 0.1V as determined from the average of 10 consecutive measurements.

4.4.6 *Rupture.*—The contacts of the relays tested as single-throw contacts shall be made to make and break 45 amp resistive load at rated voltage for 50 operations. The ON period shall be 0.2 ± 0.05 second and the OFF period shall be 30.0 seconds. No failure shall occur as a result of these tests. Following this test, the relay shall meet the tests specified in 4.4.5.

4.4.7 *Continuous Operation.*—The relays shall be subjected to room temperature (25° ± 5°C) for 100 hours with rated operating voltage across the coil and with rated continuous load through the contacts. Immediately after this these, the relay shall be subjected to the test specified in 4.4.5 and shall meet the requirements specified therein.

4.4.8 *Additional Operating Cycles.*—The contacts of the relay shall be made to make and break rated resistive load, 10 amp, at rated voltage for 25,000 operations. The ON period shall be 0.2 ± 0.05 second and the OFF period shall be 0.2 ± 0.05 second. No failure shall occur as a result of these tests. Following this test, the relay shall meet the tests specified in 4.4.5.

4.5 Qualification Testing.

4.5.1 *Qualification Test Sample.* — The qualification test sample shall consist of one panel. The sample shall be appropriately identified with the manufacturer's own part number and any additional information required by the letter of authorization.

4.5.2 *Qualification Required.* — Prior to actual procurement, the product which this specification covers shall pass the qualification tests specified herein. If the product is later modified in any way, the modified form shall be subjected to and shall pass the same qualification tests.

4.5.3 *Qualification Tests.* — Qualification tests shall consist of all tests described under 4.4.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and Packaging.

5.1.1 *Level A.*—The panel shall be preserved in accordance with MIL-P-116, method II and packed in containers conforming to JAN-P-108, or MIL-B-4229. For domestic shipment, containers conforming to LLL-B-631 or LLL-B- 636 may be used. Unit

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quantities shall be as specified by the procuring activity.

5.1.2 *Level C.*—The panel shall be preserved and packaged in accordance with manufacturer's commercial practice.

5.2 Packing.

5.2.1 *Level A.*—Panels preserved and packaged in accordance with 5.1.1 shall be packed in exterior type shipping containers conforming to PPP–B–585, PPP–B–601, JAN–P–103, JAN–P–106, JAN–P–108, or MIL–B–138. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a case liner conforming to MIL–L–10547 and shall be sealed in accordance with the appendix thereto. The case liner will not be required when the unit, intermediate, or exterior container conforms to JAN–P–108 and is sealed at all joints and seams, including manufacturer's joint, with tape conforming to JAN–P–127.

5.2.2 *Level C.*—Packages which require overpacking for acceptance by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 **Physical Protection.**—Cushioning, blocking, bracing, and bolting, as required shall be in accordance with JAN–P–100 except that for domestic shipments, waterproofing requirements for cushioning materials and containers shall be waived. Drop tests of JAN–P–100 shall be waived when preservation, packaging, and packing of the item are for immediate use or when drop tests of MIL–P–116 are applicable.

5.4 **Marking.**—Interior and exterior containers shall be marked in accordance with MIL–STD–129. The nomenclature shall be marked in accordance with MIL–STD–129. The nomenclature shall be as follows:

Panel, Control Transfer, Airfield Lighting, Type MB–1

6. NOTES

6.1 Intended Use.—The panel covered by this specification is intended for use in transferring airport lighting control circuits from the control tower to the emergency control station.

6.2 Ordering Data.—Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of preservation and packaging, and packing.

6.3 Hazards Guarantee.—The bidder shall submit proof to the procuring activity that the panel specified in 3.5 which he proposes to supply under this specification conforms to the standards of the Underwriters' Laboratories, Inc., as regards fire and casualty hazards. The label or listing of the Underwriters' Laboratories, Inc. will be accepted as evidence of conforming to the specification requirements.

6.3.1 In lieu of the label or listing, the bidder may submit independent proof satisfactory to the procuring activity that this panel conforms to the published standards, including methods of tests, of the Underwriters' Laboratories, Inc. Compliance with the fire and casualty hazards requirements does not absolve the bidder from complete compliance with the other requirements of this specification.

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6.4 Qualification.—With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date.

6.4.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the qualifying activity below:

Commander
Wright Air Development Center
ATTN: WCLE
Wright-Patterson Air Force Base, Ohio

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Preparing Activity:

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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